Supplementary material

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Description. Supporting analyses for the main text: interval recalibration, rows outside the standard analysis set, a locked device forecast, and two replications on allometric scaling laws outside fusion.

For: "Validation protocol reverses machine-learning model rankings in tokamak energy-confinement scaling" (L. Salcedo).

This document holds the supporting analyses referred to in the main text. Sections S1 to S3 are fusion analyses whose headline numbers are stated in the main text and whose construction, tables and caveats are given here in full: repairing the interval collapse and finding the limit of the repair, replicating the ranking inversion on rows the standard analysis set excludes, and a locked forecast at three device operating points. Sections S4 and S5 are the two replications outside fusion. Those two are supporting replications rather than evidence the main confinement result depends on: they show that the extrapolation failure measured on the ITPA database is not specific to tokamaks, and they probe what the ranking inversion needs and the extrapolation failure does not. Section and equation numbers prefixed S refer to this document; unprefixed section names refer to the main text.

Each section here is a result the main text states and relies on. What moved is the construction, the tables and the caveats, not the finding.

S1 Repairing the intervals, and the limit of the repair

The diagnosis in the calibrated-intervals section of the main text makes a testable prediction. If the interval collapse is really about exchangeability rather than about the models, then calibrating on held-out *labels* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. Both halves are tested below.

The label-level scheme uses an inner leave-one-label-out loop, and nothing in it touches the target label. It is called label-level rather than machine-level deliberately: the calibration unit is the database device/wall label, so for a target of JET-ILW the calibration set still contains JET. The device-level variant that removes that overlap is reported at the end of this section. For each target m: remove m; the remaining labels are the training set. Inside that training set, hold out each label k with at least 30 rows in turn, fit on the others, and record $|\log \hat{\tau} - \log \tau|$ on k together with each of its rows' Mahalanobis distance from that inner fit set. Pool those residuals into the calibration quantile. Then fit the final model on all the training labels and score m once. Every conformity score therefore comes from a label its own model had not seen, which is the unit the test rows differ by, and no row from m enters any fit or any quantile. The distance-scaled variant fits

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Table S1: Empirical coverage of nominal 90% intervals under three calibration schemes, on a held-out database label (LOLO) and across the ITER-size-matched cut. The device-level counterpart is Table S2.

	split conformal		by label		+ distance	
Model	LOLO	cut	LOLO	cut	LOLO	cut
random forest	35%	3%	**88%**	29%	88%	40%
hist grad. boosting	45%	0%	**90%**	4%	89%	13%
ridge, log-linear	83%	70%	91%	77%	92%	82%
power law, collisionless	85%	91%	91%	**95%**	92%	92%
IPB98(y,2)*	89%	88%	89%	89%	89%	87%

$\sigma(d) = \exp(c_0 + c_1 d)$ to those calibration residuals by least squares on $\log|r|$ and divides the scores by it.

Neither scheme is a finite-sample-valid group-conformal procedure, and they are not claimed as one. Two things stand in the way. The conformity score is still a row, not a label, and rows arrive in correlated clusters, so the exchangeability the split-conformal proof needs is no more available here than in the calibrated-intervals section of the main text. And in the distance-scaled arm the scale function $\sigma(d)$ is estimated from the same calibration residuals it then normalises, which the standard argument does not cover. What these are is empirical held-out-label residual calibration: the quantile is the finite-sample conformal rank applied to those scores, and the coverage reported for it is measured rather than guaranteed. Recovering a guarantee would need label-level conformity scores and a scale model fitted on separate rows or cross-fitted, and neither is done here.

Both arms behave as the diagnosis predicts. On a held-out label, empirical row-level coverage returns close to nominal: every model lands within two points of it, tree ensembles included, and the random forest goes from 35% to 88%. Across the size cut none does. Neither recalibration procedure tested restores near-nominal coverage there, because every calibration unit is smaller than every test unit, and scaling the interval by extrapolation distance recovers only part of the gap (3% to 40% for the forest). A repair that stops exactly where the diagnosis says it should is stronger evidence for the diagnosis than one that worked everywhere would be.

The same calibration over physical devices. The scheme above calibrates on database labels, so for a target of JET-ILW the calibration set still contains JET. Collapsing the wall variants and repeating the whole procedure over the 11 physical devices removes that overlap: training, calibration and scoring then all use the device as the unit.

Table S2 says the diagnosis holds and the repair is less complete than the label-level numbers suggest. Plain split conformal is worse on a genuinely unseen device than on an unseen label, 31% against 35% for the forest and 38% against 45% for the booster, which is what removing the within-device overlap should do. Calibrating by device then recovers most of the shortfall for the forest (84%, and 88% with distance scaling, against 88% either way at label level) but noticeably less for the booster, which reaches 77% and 80% where the label-level scheme reported 90% and 89%. The label-level number therefore overstates how well this repair calibrates the booster on a device it has never seen. Both linear models are unaffected within a point or two. The cost is visible in the widths: the trees buy that coverage at roughly 2.3 to 2.8× multiplicative half-width against 1.23× under plain split

Table S2: The same three schemes with wall variants collapsed, so every calibration unit is a physical device and the held-out device is genuinely unseen. Pooled row-level coverage of nominal 90% intervals, with the median multiplicative half-width in brackets.

Model	split conformal	by device	+ distance
random forest	31% (1.23×)	84% (2.46×)	88% (2.76×)
hist grad. boosting	38% (1.24×)	77% (2.29×)	80% (2.53×)
ridge, log-linear	85% (1.34×)	92% (1.40×)	93% (1.43×)
power law, collisionless	88% (1.34×)	92% (1.39×)	93% (1.42×)
IPB98(y,2)*	90% (1.39×)	90% (1.39×)	90% (1.39×)

*historical reference, not refitted within these folds.

conformal, where the linear models stay near 1.4×.

Across the size cut with the same device grouping the picture is unchanged in kind. The forest and booster remain far from nominal at 41% and 12% by device and 57% and 21% with distance scaling, better than the label-level 29% and 4% but nowhere near 90%, while the constrained power law holds at 95% and 93%. Changing the calibration unit cannot repair a shift the calibration set does not contain, which is the point of the section.

Among the models refitted within folds, the constrained power law of the constraint section of the main text is the only one in Table S1 whose plain split-conformal coverage stays near nominal at the ITER-size-matched cut, and its intervals hold there under every scheme. IPB98(y,2) also holds at 88%, but it is a historical reference rather than a fold-fitted competitor.

S2 Robustness on rows the standard analysis set excludes

Everything above rests on one file, which is a central limitation of the preceding analysis. The same OSF project publishes the full database revision, DB5.2.3, with 14153 rows against STD5's 6228, and we pin it by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it and leaves two populations this study had not analysed: **5358 H-mode rows STD5 does not contain** (12 labels, zero row overlap) and **3860 ohmic and L-mode rows** in a different confinement regime, which we score against ITER89-P [3] because an H-mode law is the wrong baseline for an L-mode plasma.

This replication has one failure mode that would not announce itself. If the row match against STD5 fails, every STD5 row stays in the "disjoint" arm, that arm becomes a superset of the data the ranking-inversion section of the main text was computed on, and it reproduces the ranking-inversion section of the main text by construction. The matcher is therefore checked positively as well as negatively: STD5 matched against itself must overlap completely, and the disjoint arm must share zero rows.

The ranking inverts in both arms. On the disjoint H-mode rows the best cross-validated model beats the published law by 42%, a margin of the same size as the 36% over that law reported above and computed on rows it never saw, and the best model on an unseen label is IPB98(y,2). On the ohmic and L-mode rows the cross-validation gain over ITER89-P is 67% and the best model on an unseen label is the collisionless power law of the constraint section of the main text. Counting label–model pairs where a tree ensemble loses to the published law on an unseen label: 19 of 24 and 10 of 10.

Row disjointness is not discharge disjointness, and the paper's own argument says the discharge is the unit that matters. Of the 5358 H-mode rows, 425 (7.9%) are slices of one of 336 discharges that STD5 samples at a different time. Dropping those discharges outright leaves 4933 rows across 2864 discharges that share no shot with STD5 at all, and the arm survives: the best cross-validated model still beats the published law, by 39% against 41% on the full arm, and the best model on an unseen label is still IPB98(y,2), with the tree ensembles at 0.526 and 0.552 against the power law's 0.332. The reversal does not depend on the shared discharges.

So the reversal is not an artifact of the standard set's selection criteria, and not a property of ELMy H-mode or of IPB98(y,2) specifically. It is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

S3 Locked predictions at three device operating points without matching H-mode observations

Everything above is retrospective. To make the disagreement checkable rather than merely described, we record what each model predicts for three real devices, with intervals, a date, and a digest over the input rows so that a later edit leaves a mark. The parameter sets are published design values, and for SPARC and ITER, evaluating IPB98(y,2) on the adopted design vector reproduces the reference confinement time its own source quotes, to 0.6% and 2.9% respectively; the JT-60SA source supplies operating parameters rather than a quoted confinement time. Those are checks that the right machine is being described. None of the three has a *measured* H-mode thermal confinement time at the operating point tabulated here, which is what makes these predictions rather than retrodictions.

Table S3: Predicted energy confinement time, in seconds, with nominal 90% intervals. Locked 31 August 2026 against a fixed dataset digest. Operating points are published design values: SPARC from [2] (V2 primary reference discharge), ITER from [1] ($Q = 10$ inductive baseline), and JT-60SA from its Research Plan [4] (full-current inductive scenario). Intervals are distance-scaled held-out-label calibration intervals, not the plain split-conformal intervals of the main text's coverage table, and are pointwise intervals for a new observation; their construction is specified in the text below. The complete input vector for each device is recorded in `results/forecast.json` under the content digest quoted below.

	SPARC 1.85 m	JT-60SA 2.96 m (operating)	ITER 6.2 m
IPB98(y,2)*	0.765 [0.58, 1.00]	0.479 [0.35, 0.66]	**3.591 [2.66, 4.85]**
power law, collisionless	0.724 [0.55, 0.96]	0.428 [0.31, 0.59]	2.837 [2.09, 3.84]
ridge, log-linear	0.720 [0.53, 0.98]	0.444 [0.32, 0.62]	2.858 [2.07, 3.95]
random forest	0.136 [0.03, 0.53]	0.449 [0.23, 0.87]	**0.435 [0.19, 0.98]**
hist grad. boosting	0.305 [0.14, 0.69]	0.418 [0.24, 0.74]	0.444 [0.24, 0.84]

How these intervals are built. They are not the plain split-conformal intervals of the calibrated-intervals section of the main text, and deliberately so: the main text's coverage table measures those covering 3% of rows across the ITER-size-matched cut, so quoting them for a next-step device would publish an interval already known to fail in exactly this situation. Each row instead uses the distance-scaled label-level scheme of Sec. S1, which gives the best size-cut calibration for the models that fail most severely there, applied with the whole database as the training set. Two disclosures belong with that. The scheme was chosen after comparing the three in Table S1, so these intervals are exploratory rather than prospectively selected, in the same sense as the constraint rung of the constraint section of the main text. And it calibrates over database labels rather than physical devices, so these are not device-level conformal intervals: Table S2 measures what that distinction costs, and for the tree ensembles it is not negligible. The label-level scheme is retained here because it is the one the forecast was locked under on 31 August 2026, and re-deriving the intervals afterwards would make the lock meaningless. A reader who prefers the device-level calibration should read the tree rows of Table S3 as correspondingly optimistic in width. For a given model, each eligible label k is held out in turn, the model is refitted on the remaining labels, and the absolute log residuals on k are recorded together with each of its rows' Mahalanobis distance d from that inner fit set. A scale $\sigma(d) = \exp(c_0 + c_1 d)$ is fitted to those pairs by least squares on $\log|r|$, dropping exactly-zero residuals; the conformity scores are the residuals divided by $\sigma(d)$; and the half-width is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest such score at $\alpha = 0.10$, the same finite-sample rank used in the calibrated-intervals section of the main text. The interval reported for a device is that quantile multiplied by $\sigma(d)$ evaluated at the device's own distance, applied symmetrically in $\log \tau$ and exponentiated. The model whose point prediction is tabulated is then fitted once on all 6228 rows. For IPB98(y,2) the same construction is used on its analytic residuals, so its interval is comparable to the others rather than merely coexisting with them. Every interval is therefore an empirical pointwise calibration interval for a new observation rather than a finite-sample-valid conformal interval. Three things stand in the way, and only the last is about ITER: the conformity scores remain clustered row-level scores, the distance scale is estimated from the calibration residuals it normalises, and every calibration unit is smaller than ITER along the size direction, which is the limitation Sec. S1 measures directly.

Table S3 illustrates the practical consequence of the disagreement in one place. On JT-60SA, which sits inside the database's size range and is already operating, all five models agree to within 15%. On ITER, 1.82× beyond it, they disagree by a factor of **8.3**.

For the random forest that gap cannot be closed by tuning the standard estimator, and the ranking-inversion section of the main text says why: it predicts by averaging training targets, so its output cannot exceed the largest one, which here is 1.321 s, and no hyperparameter moves that ceiling. The gradient booster carries no corresponding arithmetic bound but produces a similarly low point prediction. Because its prediction cannot exceed the largest training target, 1.321 s, the random forest cannot reach the 2.84 s to 3.59 s range the linear and physics-informed models predict for this ITER operating point, and its nominal 90% interval there runs from 0.19 s to 0.98 s, containing none of those predictions.

The JT-60SA column becomes checkable first, though not against any measurement published as of the 31 August 2026 forecast lock: the first campaign ran L-mode divertor plasmas, and its published power balance reports the energy confinement time following L-mode scaling [5], whereas the row above is an H-mode prediction scored against IPB98(y,2). Checking it needs H-mode operation at a comparable design point, which the research plan [4] places in a later campaign and which the overview of first operation and forward plans [6] discusses in view of ITER and DEMO; we deliberately do not put a date on that here, because the schedule is not ours to assert and the prediction is locked against a parameter vector rather than against a calendar.

Each row of Table S3 carries its own interval and is assessed separately against it, so the range quoted here names one model rather than the set. A stationary JT-60SA H-mode near this design point reporting a thermal confinement time outside 0.349 s to 0.657 s would fall outside the IPB98(y,2) reference row's stated 90% predictive interval. The fitted rows are assessed at their own bounds, which are narrower for the constrained power law (0.313 s to 0.587 s) and wider for the random forest (0.232 s to 0.867 s). A single observation outside a nominal 90% interval is evidence against that forecast rather than a refutation of it: a perfectly calibrated model still places about one observation in ten outside such an interval, so a single miss is inconsistent with the prediction at the stated interval level and no more. Because all five intervals overlap across roughly this range, a measurement inside it separates none of them: this row tests whether any model is right, not which, and that is why the ITER column carries the argument.

Nothing above anticipated one feature of this table: SPARC sits further from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine. Its 12.2 T field is 2.1 times the largest toroidal field anywhere in the cleaned STD5 rows, 5.82 T on Alcator C-Mod, and every other label in the database tops out at or below 4.80 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

S4 The same audit on a scaling law from another science

Every result in the main text rests on ITPA data, which leaves open whether any of it is a fact about fusion specifically. Mammalian basal metabolic rate against body mass is the same object and the older one: Kleiber [7] found in 1932 that metabolic rate scales as mass to the 3/4. Whether the true exponent is 3/4 or 2/3 has been argued for decades and we take no position; 3/4 is used here as the canonical historical reference law a fitted model has to beat. Taxonomic *order* is the unit a species belongs to, as a tokamak is the unit a discharge belongs to, and *body mass* is the axis a next instance lies beyond, as machine size is. The data is Supplement 1 of a metabolic-rate compilation deposited on Figshare (article 3549807), pinned by SHA-256 as the HDB5 files are, giving 541 species records across 11 orders whose median masses span 342x. The analysis runs through the domain-agnostic module released with this work rather than a copy of the fusion pipeline, so it is the procedure a reader would apply to their own data, and the problem has a *single predictor*: no model can win by finding a better combination of features, so the only thing separating them is functional form. Refitting the exponent freely gives 0.687 against Kleiber's 0.75, so the constraint is not free here either.

Table S4 reports the three splits and Figure S1 shows them.

The extrapolation failure reproduces. Both tree ensembles are worse than both power laws at all 8 mass cuts, and the random forest loses to Kleiber on 9 of 11 held-out orders. Two signatures reproduce with it. Tree error rises with extrapolation distance, at $\rho = +0.64$ and $+0.65$ against $+0.39$ and $+0.45$ for the power laws, which is the same sign as the distance diagnostic in the main text but a much narrower contrast, since here the power laws rise with distance too. And the training-range bound binds at every cut.

Table S4: Kleiber's law under the same three splits. Scores are log-RMSE.

Model	CV, by species	leave-one-order-out	widest mass cut
Kleiber, exponent 3/4	0.396	0.440	**0.374**
power law, free exponent	0.374	0.432	0.496
hist grad. boosting	0.418	0.487	0.889
random forest	0.437	0.517	0.710

The ranking reversal does not, and that bounds the claim. The trees never win the easy split either: under cross-validation by species they score 0.437 and 0.418 against the free power law's 0.374. With no inflated margin there is nothing available to invert. The HDB5 gain came from a flexible model exploiting structure across nine features; with one predictor and a relationship close to a straight line in logs, a tree has far less to exploit and does not buy the interpolation win that later reverses. So the inversion tracks whether the flexible model wins interpolation first, which no single database could have shown. Whether predictor count is itself the operative variable is a separate question this comparison does not settle.

A limitation of this replication. The 11 taxonomic orders are not statistically independent: they share phylogeny, so two orders that diverged recently are more alike than two that did not, and the held-out group is therefore not a draw from a pool of independent units. Comparative biology normally handles this with phylogenetically controlled regression [10]; this audit does not. It treats orders only as held-out groups, and models no phylogenetic covariance. What that costs is any rate quoted here: the counts below establish a direction, not a frequency with which a constraint helps.

The constraint result reproduces in direction, not in strength. Kleiber's published exponent costs +0.023 under cross-validation and wins the widest cut, 0.374 against 0.496, where the held-out order is 69x heavier than the training median, which is exactly the trade the Connor–Taylor constraints make in the main text. But it is weaker, winning at 4 of 8 cuts rather than 8 of 8, decisively at the extremes and narrowly losing in the middle. That a constraint helps out of distribution replicates; that it helps reliably does not.

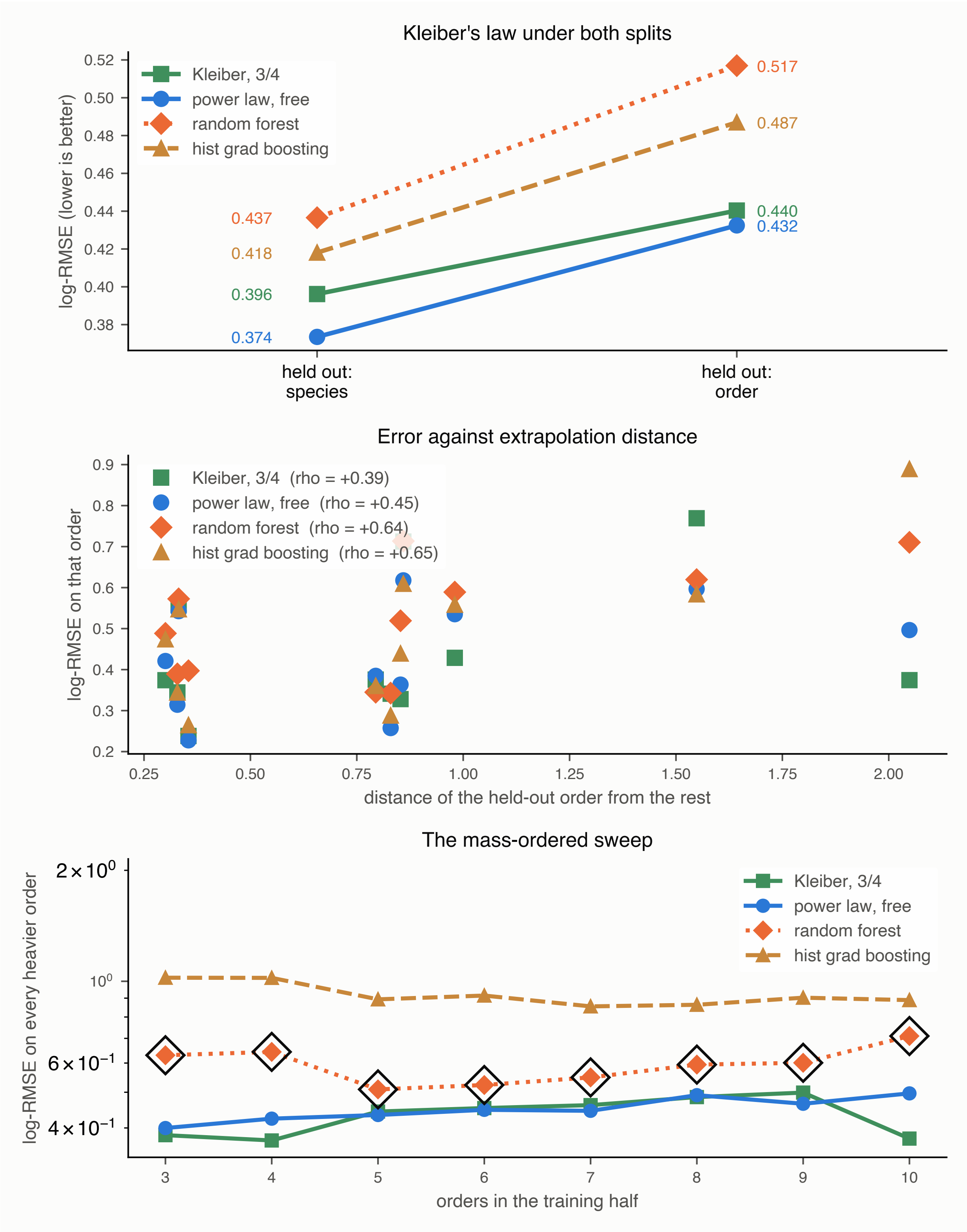

Figure S1: Kleiber's law under the same three splits. Top: no inversion, because the trees lose under both splits and so have no interpolation margin to give up. Middle: error against extrapolation distance, rising for the flexible models as it does on the ITPA H-mode database. Bottom: the mass-ordered sweep, with the circled points marking cuts where the random forest is held inside its training range.

S5 The reversal's precondition, measured directly

Sec. S4 leaves an asymmetry. The extrapolation failure reproduced in full on Kleiber's law; the ranking inversion did not, and the explanation offered there was a conjecture: with a single predictor a tree has little to exploit, never wins the interpolation split, and so has no advantage to lose. Two datasets cannot test that, because HDB5 and the mammalian compilation differ in feature count and also in science, group structure, sample size and noise.

A third dataset can probe that with a nested predictor ladder on the same rows and splits. The Biomass And Allometry Database [8] records, for individual plants, a nested set of measurements: basal diameter, height, leaf area and leaf mass. Taking the rows complete in all four gives 3599 plants across 53 species, 33 of them with the 30 individuals we require to hold a species out, spanning 36x in diameter across species medians and over seven orders of magnitude in mass. It is released CC0 and pinned by SHA-256 as the other deposits are. Species plays the part of tokamak, diameter the part of machine size, and the branching-network prediction that mass scales as diameter to the 8/3 [9] the part of IPB98(y,2): an exponent derived from theory rather than fitted, which refits here to 2.512 and costs 0.5641 against 0.5780 in sample when held.

Each rung of the ladder is a prefix of the next and all four are scored on the same rows. The rows, the splits and the model specifications are held fixed; each rung adds one named predictor to the nested feature set. Predictor count is therefore what is varied, but not the only thing that changes with it, since each added column carries its own biological information. The split structure matters and is easy to get wrong: grouped cross-validation by discharge keeps every machine on both sides, so its analogue keeps every species on both sides. Grouping the cross-validation by species instead would be a second leave-one-group-out, and against it no reversal can appear at any dimension.

Table S5 and Figure S2 separate two statements that had been travelling as one. The interpolation margin rises monotonically and crosses zero between two and three predictors. The extrapolation margin does not shrink as the trees improve; it widens, and the power law wins the held-out species at 4 of 4 rungs. The inversion therefore appears at 3 predictors and every rung above, which is exactly where the interpolation margin turns positive. What the ladder shows is that the inversion tracks the interpolation margin, not that three is a threshold: each new column adds real biological information as well as a dimension, and leaf mass in particular is closely related to the target, so the rungs differ in more than their count.

So the extrapolation failure reproduces in all three datasets tested here, at every predictor count tested. In this ladder the extrapolation disadvantage persists at every rung, while the ranking inversion appears only once the flexible model gains an interpolation advantage. The inversion is therefore not necessary for the extrapolation failure, and is better read as a visible symptom of it than as the thing itself. At the one- and two-predictor rungs the same qualitative extrapolation hazard is present without an interpolation advantage whose loss would signal it.

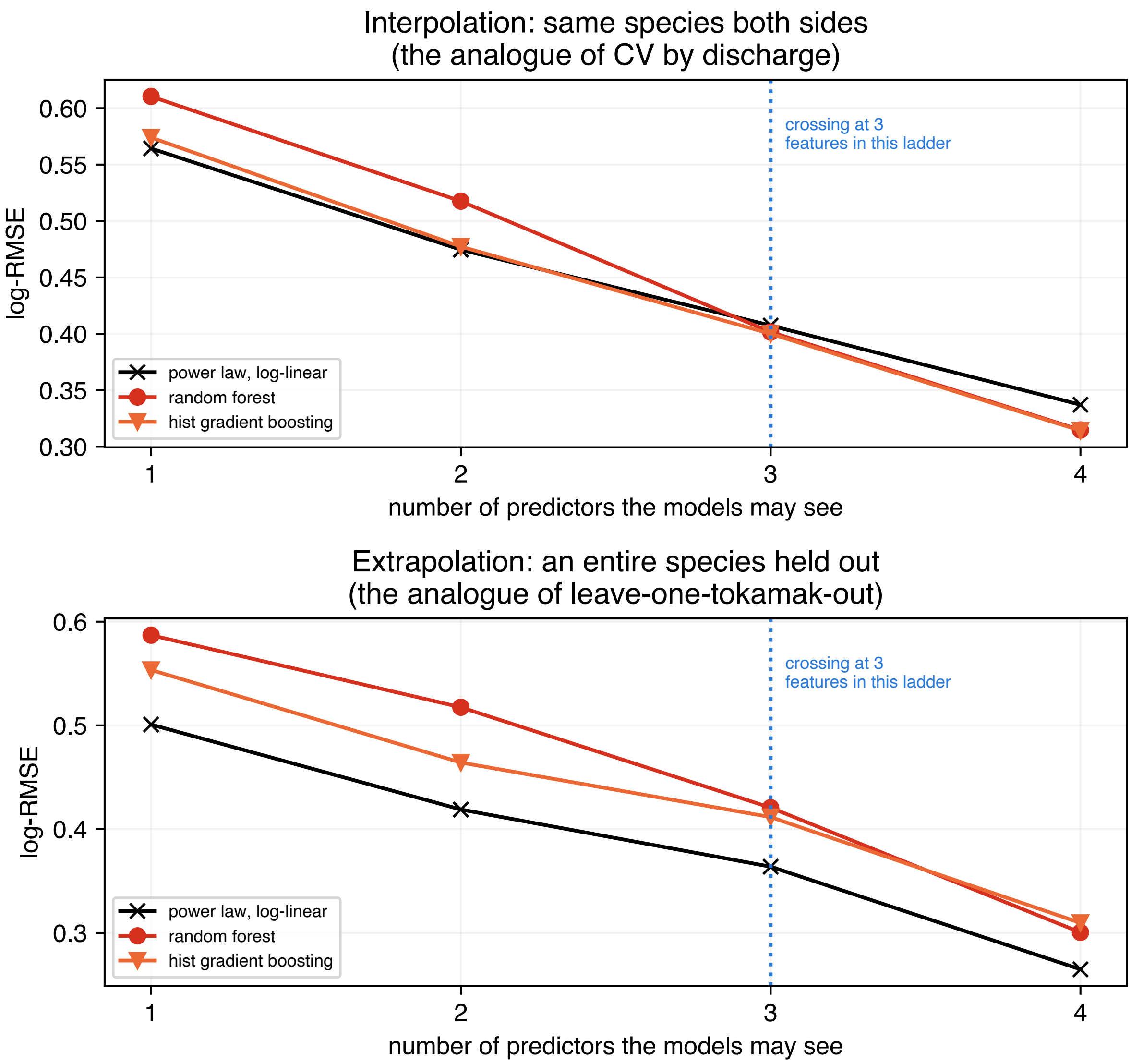

Figure S2: The reversal's precondition, on 3599 plants held fixed. Top: the interpolation split, where every species appears on both sides. Bottom: an entire species held out. The rows, splits and model specifications are held fixed; each rung adds one named predictor to the nested set. The top curves cross; the bottom ones do not.

predictors	interp. gain	held-out gain	reversal
diameter	−1.7%	−10.5%	no
+ height	−0.6%	−10.8%	no
+ leaf area	+1.8%	−13.1%	yes
+ leaf mass	+6.8%	−13.4%	yes

Table S5: The best tree ensemble's margin over the log-linear power law, on 3599 plants held fixed across the four rungs. Positive favours the trees. The interpolation column crosses zero; the extrapolation column does not, and grows slightly against them.

S6 Full model, kernel and split specification

This section carries the settings the main text points at. "Library defaults" is not a specification, and a default may differ in a later release, so the consequential settings are written out rather than referenced. Library versions are pinned in the repository's `constraints.txt` and named in the main text.

S6.1 The models of the headline comparison

The power law is `StandardScaler` followed by `Ridge` with $\alpha = 1$ and the SVD solver, chosen before any held-out score was computed and never varied. The random forest is `RandomForestRegressor` with 300 trees, `criterion="squared_error"`, unlimited depth, `min_samples_split` = 2, `min_samples_leaf` = 1, `max_features` = 1.0 and bootstrap resampling. The gradient booster is `HistGradientBoostingRegressor` with squared-error loss, learning rate 0.1, at most 100 boosting iterations, 31 leaves, no depth limit, `min_samples_leaf` = 20, no L_2 penalty and 255 bins. Its `early_stopping="auto"` resolves to *off* at every sample size here, since that setting enables early stopping only above 10 000 rows and no training fold reaches that. The polynomial controls are `PolynomialFeatures` of degree 2 and 3 without a bias column, then the same scaler and ridge. Neither tree ensemble is standardised, which is immaterial to both.

S6.2 The Gaussian-process kernels

The three kernels are

$$\begin{aligned} k_{\mathrm{RBF}}(x,x') &= \sigma_r^2 \exp\Big(-\tfrac{1}{2}\|x-x'\|^2/\ell^2\Big) + \sigma_n^2\delta_{xx'}, \\ k_{\mathrm{lin}}(x,x') &= \sigma_l^2\,(x^\top x' + \sigma_0^2) + \sigma_n^2\delta_{xx'}, \\ k_{\mathrm{lin+RBF}}(x,x') &= k_{\mathrm{lin}} + k_{\mathrm{RBF}} - \sigma_n^2\delta_{xx'}, \end{aligned} \tag{1}$$

with a single isotropic length scale ℓ shared across the standardised log features, so the RBF component is isotropic rather than automatic-relevance-determining. Features are standardised inside the pipeline, so each fold's scaler is fitted on that fold's training rows alone and ℓ is in those units. The prior mean is constant and `normalize_y` is on, which puts it at the training mean of $\log\tau$: this is what makes "reverts to the prior" and "returns the average training target" the same statement, and the parallel with a tree ensemble exact. Observation noise enters as an additive `WhiteKernel` rather than through a fixed jitter. Hyperparameters are fitted by maximising the log marginal likelihood with scikit-learn's default L-BFGS-B optimizer, `n_restarts_optimizer` = 0, from the shared initialisation $\sigma_l^2 = 1$, $\sigma_0 = 1$, $\sigma_r^2 = 0.5$, $\ell = 1$, $\sigma_n^2 = 0.05$, under the bounds $\sigma_l^2, \sigma_0 \in [10^{-3}, 10^3]$,

$\sigma_r^2 \in [10^{-3}, 10^3]$, $\ell \in [10^{-2}, 10^3]$ and $\sigma_n^2 \in [10^{-6}, 1]$. Every rung starts from the same values for every term it has, so the three fits differ only in which terms exist. The optimisation runs on a subsample of exactly 1000 training rows drawn without replacement by `numpy.random.default_rng(42)`, or on the whole fold when it holds fewer; the kernel is then frozen and the exact GP is conditioned on *every* training row of that fold, which is one Cholesky factorisation. No held-out row enters either stage. Hand-chosen hyperparameters are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure. Intervals quoted for these models are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term, exponentiated to τ.

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